Feasibility Study of PeerGrid: A Kerala-Focused College Connectivity and Recommendation Platform

[TEAM 2]

Abstract*— *This paper presents a feasibility study for PeerGrid, a proposed Kerala-focused college connectivity platform that combines KEAM rank-based college recommendations, verified student networking, college rankings, campus event management, and authentic student reviews. The study evaluates the project from four perspectives — technical, economic, operational, and schedule feasibility — to determine whether the system can be practically developed, deployed, and adopted. Results indicate that PeerGrid is highly feasible across all four dimensions, relying on proven open-source technologies, a low development cost, strong alignment with real user needs, and a realistic sixteen-week Agile development schedule.

Index Terms*— *feasibility study, college recommendation system, KEAM, student networking platform, mobile application, software engineering.

I. INTRODUCTION

A feasibility study is an important phase in software engineering that evaluates whether a proposed system can be successfully developed and implemented. It helps determine whether the project is practical, financially justifiable, technically achievable, operationally acceptable, and capable of being completed within the planned timeframe. Conducting a feasibility study reduces project risks by identifying potential challenges before development begins.

For the proposed PeerGrid platform, the feasibility study was carried out from four major perspectives: technical feasibility, economic feasibility, operational feasibility, and schedule feasibility. PeerGrid is intended to serve as a Kerala-focused college connectivity platform that combines KEAM rank-based college recommendations, verified student networking, college rankings, campus event management, and authentic student reviews. The following sections evaluate the practicality and viability of the project from each perspective.

II. TECHNICAL FEASIBILITY

Technical feasibility examines whether the required technology, software tools, hardware resources, and technical expertise are available for developing the proposed system. PeerGrid is designed using a modern technology stack consisting of Flutter for cross-platform mobile application development, React.js for the administrative web dashboard, Node.js with Express.js for backend services, PostgreSQL and MongoDB for structured and unstructured data storage, and Python with machine learning libraries such as Scikit-learn, Pandas, and NumPy for the recommendation engine.

These technologies are well established, open source, and widely adopted in industry, making them reliable for academic as well as commercial software development. The development environment can be set up using standard personal computers without requiring specialized hardware. Cloud platforms such as AWS, Firebase, or Google Cloud provide scalable hosting, authentication, database services, push notifications, and backup facilities, reducing infrastructure complexity.

The recommendation engine can be trained using historical KEAM allotment trends and college data, while REST APIs and JWT authentication enable secure communication between the client and server. Since the required programming languages, frameworks, and cloud services are readily available and the development team has the necessary knowledge of mobile application development, databases, web technologies, and software engineering principles, the proposed system is technically feasible.

Furthermore, the modular architecture of PeerGrid allows future enhancements such as AI-powered counselling, alumni mentoring, scholarship recommendations, and expansion to other entrance examinations without redesigning the complete system. This demonstrates that the project is not only technically achievable but also scalable and maintainable.

III. ECONOMIC FEASIBILITY

Economic feasibility evaluates whether the expected benefits of the proposed system justify the overall development and operational costs. One of the major advantages of PeerGrid is that it can be developed primarily using open-source technologies. Flutter, React.js, Node.js, PostgreSQL, MongoDB, Python, GitHub, and most development tools are freely available, significantly reducing software licensing costs.

During the development phase, educational or free-tier cloud services can be used for hosting, authentication, database storage, and notifications. The primary expenses include internet connectivity, domain registration, optional cloud hosting, testing devices, and deployment charges for mobile application stores. These costs are relatively small compared to the value delivered by the platform.

The long-term benefits of PeerGrid include helping students make informed college choices, reducing misinformation during KEAM counselling, improving communication between prospective students and current students, and providing colleges with an additional platform to showcase academic achievements and campus events. In the future, the platform may also generate revenue through premium analytics, institutional partnerships, sponsored educational events,

verified promotional listings, or placement-related services. Since the anticipated benefits considerably outweigh the expected development and maintenance costs, PeerGrid is economically viable and financially feasible.

IV. OPERATIONAL FEASIBILITY

Operational feasibility determines whether the proposed system can function effectively in the intended environment and whether users are willing to adopt it. PeerGrid has been designed by considering the requirements of prospective students, current students, alumni, and administrators involved in Kerala's higher education ecosystem.

Students frequently face difficulties in selecting colleges because information is scattered across government websites, social media groups, coaching centres, and informal discussions. PeerGrid solves this problem by integrating verified college information, historical admission trends, authentic student reviews, social networking, and campus event updates into a single platform. The user interface is designed to be simple, responsive, and mobile-friendly so that even first-time users can navigate the application with minimal training.

The administrator module provides facilities for user verification, content moderation, college profile management, and report generation, ensuring that the platform remains reliable and trustworthy. Since the application directly addresses existing problems experienced during the KEAM admission process and offers practical benefits to students, colleges, and administrators, user acceptance is expected to be high. Therefore, the proposed system is operationally feasible.

V. SCHEDULE FEASIBILITY

Schedule feasibility assesses whether the proposed project can be completed within the available time and resources. Based on the project implementation plan, PeerGrid can be developed over approximately sixteen weeks using an Agile software development methodology. The project activities include requirement analysis, literature survey, system design, database design, data collection, backend development, machine learning model development, frontend implementation, integration testing, user acceptance testing, and beta deployment.

Several development activities can be carried out simultaneously. For example, while one team member develops backend APIs, another can design the Flutter application and another can prepare datasets and recommendation models. This parallel development approach reduces project duration and improves productivity. Continuous integration, version control through GitHub, and periodic testing further reduce development risks.

The estimated schedule provides sufficient time for debugging, performance optimization, documentation, and user feedback before deployment. Considering the available manpower, project scope, and implementation methodology, the proposed development timeline is realistic and achievable. Therefore, PeerGrid is schedule feasible.

VI. OVERALL FEASIBILITY ASSESSMENT

Table I summarizes the outcome of the feasibility analysis across all four dimensions.

TABLE I
OVERALL FEASIBILITY ASSESSMENT

Feasibility	Result	Justification
Technical	Highly Feasible	Uses proven open-source technologies, cloud infrastructure, and scalable architecture.
Economic	Highly Feasible	Low development cost with significant educational and long-term business value.
Operational	Highly Feasible	Addresses real user needs and integrates naturally with the KEAM admission process.
Schedule	Highly Feasible	Can be completed within the planned 16-week schedule using Agile development.

VII. CONCLUSION

The feasibility analysis confirms that PeerGrid is a practical and viable software project. From a technical perspective, the required technologies, development tools, and infrastructure are readily available. Economically, the use of open-source software and cloud services keeps development costs low while providing substantial value to students and educational institutions. Operationally, the system addresses genuine problems faced during the KEAM admission process and is expected to achieve strong user acceptance. Finally, the proposed implementation schedule is realistic and can be completed within the planned timeframe. Based on these findings, the development of PeerGrid is considered feasible and suitable for implementation.

REFERENCES

[1] I. Sommerville, Software Engineering, 10th ed. Boston, MA: Pearson, 2016.

[2] KEAM Office, "Kerala Engineering Architecture Medical Entrance Examination Prospectus," Commissioner for Entrance Examinations, Kerala, 2024.

[3] Author, "Add remaining references in IEEE format," [Add source], [Year].